

Useful Formulas

1 Accumulation Functions & Interest rate conversions

1.1 Accumulation Functions

- (1.) Effective interest rate from t_1 to t_2 : $i_{[t_1, t_2]} = \frac{a(t_2) - a(t_1)}{a(t_1)}$
- (2.) Effective discount rate from t_1 to t_2 : $d_{[t_1, t_2]} = \frac{a(t_2) - a(t_1)}{a(t_2)}$
- (3.) Simple Interest accumulation function: $a(t) = 1 + rt$ (where $r = i_{[0,1]}$)
- (4.) Simple Discount accumulation function: $a(t) = \frac{1}{1 - rt}$ (where $r = d_{[0,1]}$)
- (4.) Compound Interest accumulation function: $a(t) = (1 + i)^t$ (where $i = i_{[t, t+1]}$ for any t)
- (5.) Discount function: $v(t) = \frac{1}{a(t)}$ and for compound interest $v = v(1) = \frac{1}{1 + i}$

1.2 Compound Interest Rate Conversions

- (1.) $1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$ where $i^{(m)}$ is the nominal interest rate compounded m times per year
- (2.) $1 - d = \left(1 - \frac{d^{(m)}}{m}\right)^m$ where $d^{(m)}$ is the nominal discount rate compounded m times per year
- (3.) $d = \frac{i}{1 + i}$ and $i = \frac{d}{1 - d} = \frac{d}{v}$
- (4.) Constant force of interest or nominal rate convertible continuously:

$$\delta = \ln(1 + i) \iff e^\delta = 1 + i$$

- (5.) Force of interest at time t : $\delta_t = \frac{a'(t)}{a(t)} = \frac{d}{dt} (\ln(a(t))) \iff a(t) = e^{\int_0^t \delta_r dr}$

2 Annuity Formulas

(1.) Present value of an annuity immediate (value at time $t = 0$ of payments of \$1 at times $t = 1, 2, \dots, n$)

$$a_{\overline{n}|i} = \frac{1 - v^n}{i}$$

(2.) Value at time t of payments of \$1 at times $t = 1, 2, \dots, n$: $a_{\overline{n}|i}(1+i)^t$. When $t = n$:

$$s_{\overline{n}|i} = a_{\overline{n}|i}(1+i)^n$$

(3.) Present value of an annuity due: $\ddot{a}_{\overline{n}|i} = a_{\overline{n}|i}(1+i)$ but if you insist on knowing the formula we have

$$\ddot{a}_{\overline{n}|i} = \frac{1 - v^n}{d}$$

(4.) Perpetuities: $a_{\infty|i} = \frac{1}{i}$ and $\ddot{a}_{\infty|i} = \frac{1}{d}$

(5.) Annuity with geometric payments (growth rate g):

$$PV = \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g}$$

(6.) Annuity with arithmetic payments (starting payment P and additive growth Q)

$$(I_{P,Q}a)_{\overline{n}|i} = Pa_{\overline{n}|i} + \frac{Q}{i}(a_{\overline{n}|i} - nv^n)$$

(7.) Perpetuity with increasing payments:

- Geometric progression: $\frac{1}{i - g}$
- Arithmetic progression: $\frac{P}{i} + \frac{Q}{i^2}$

(8.) Annuities with payments less frequent (k times with effective rate I) than the given interest period (with rate i)

$$a_{\overline{n}|I} = \frac{a_{\overline{n}|i}}{s_{\overline{k}|i}}$$

(8.) Annuities with payments more frequent than the given interest period (with rate i)

$$a_{\overline{n}|i}^{(m)} = a_{\overline{n}|i} \cdot \frac{i}{i^{(m)}} \quad \text{and} \quad \ddot{a}_{\overline{n}|i}^{(m)} = \ddot{a}_{\overline{n}|i} \cdot \frac{i}{d^{(m)}}$$

(9.) Continuously paying annuities at a rate of $f(t)$: $\bar{a}_{\overline{n}|i} = \int_0^n f(t)v(t)dt$

3 Loans

(1.) Outstanding loan balance at time t (directly after any payment at time t) of a loan L if every payment is *exactly* P

- Retrospective: $B(t) = L(1+i)^t - Ps_{\overline{t}|i}$ (Loan moved forward – FV of payments made)
- Prospective: $B(t) = Pa_{\overline{n-t}|i}$ (PV of future payments)

(2.) Recursive outstanding loan balances at time: $B(t+1) = B(t)(1+i) - \text{Pmt}(t)$

(3.) Interest in payment at time t : $\text{In}(t) = iB(t-1)$

(4.) Principal in payment at time t : $\text{Pr}(t) = \text{Pmt}(t) - \text{In}(t)$

(5.) Principal paid from time t_1 to time t_2 : $B(t_1) - B(t_2)$

(6.) Interest paid from time t_1 to time t_2 : $\sum_{t=t_1}^{t_2} \text{Pmt}(t) - (B(t_1) - B(t_2))$

4 Bond Formulas

(1.) Coupon amounts: $Fr = Cg = Gj$

(2.) The price of a bond: $P = Fra_{\overline{n}|j} + Cv_j^n$

(3.) The premium-discount formula for a bond:

- Premium ($P > C$) = $P - C = C(g-j)a_{\overline{n}|j}$
- Discount ($C > P$) = $C - P = C(j-g)a_{\overline{n}|j}$

(4.) Book value at time t using premium-discount formulas:

$$B(t) = C(g-j)a_{\overline{n-t}|j} + C$$

(5.) Interest in coupon at time t :

$$\text{In}(t) = Cg - C(g-j)v_j^{n-t+1}$$

(6.) Amortization of premium or Accumulation of discount at time t :

$$\text{Pr}(t) = C(g-j)v_j^{n-t+1}$$